

# Project File Guide

What every file and folder in the project is, and what it does



|                   |                                                          |
|-------------------|----------------------------------------------------------|
| Project           | AI-Based Predictive Maintenance for CNC Milling Machines |
| Monitored machine | CNC Milling Machine                                      |
| Files described   | 196 project files in 10 folders                          |
| Generated         | 30 September 2026                                        |

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# 1. Folder map

The project is split by job: the data-and-model pipeline (src/), the server (backend/), the web screens (frontend/), and their inputs and outputs. Sizes are measured on disk when this document is generated.

| Folder         | Files | Size     | Contents                                  |
|----------------|-------|----------|-------------------------------------------|
| (root)         | 9     | 621.5 KB | Project root                              |
| src/           | 14    | 93.8 KB  | data and model pipeline (Phase 1)         |
| backend/       | 14    | 114.4 KB | engine and API (Phase 2)                  |
| frontend/      | 18    | 208.9 KB | the web interface (Phase 3)               |
| data/          | 3     | 8.4 MB   | datasets and the live database            |
| models/        | 2     | 21.6 MB  | trained model                             |
| artifacts/     | 19    | 2.4 MB   | figures and metrics                       |
| docs/          | 35    | 17.1 MB  | documents and their generators            |
| tests/         | 5     | 33.5 KB  | automated tests                           |
| batch_testing/ | 77    | 2.4 MB   | Excel files for the Batch Analysis screen |

## Where to start

To run the app: `.venv\Scripts\python.exe -m backend.app`, then open `http://127.0.0.1:8000/`. To read about the project: `README.md`, then `docs/02_FINAL_YEAR_PROJECT_REPORT.pdf`. To test it quickly: `docs/ESSENTIAL_WEBSITE_TEST_CASES.md`.

## 2. Project root

The top-level files. Open README.md first.

| File                    | Size     | What it is                                                                                                                                                                                                                    |
|-------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| .gitignore              | 874 B    | Tells Git which files not to track: the virtual environment, caches, the downloaded/expanded datasets, trained models and the runtime database.                                                                               |
| .python-version         | 5 B      | Tells Render (and pyenv) to use Python 3.12.                                                                                                                                                                                  |
| generate_batch_tests.py | 31.4 KB  | Rebuilds the whole batch_testing/ folder: creates every test Excel file, uploads each one to the real API and writes 00_TEST_INDEX.xlsx with what the app returned.                                                           |
| Main synopsis (2).pdf   | 556.6 KB | The original project synopsis (19 pages): objectives, block diagram, methodology, hardware/software list and the 10 literature reviews. The source of the project's requirements.                                             |
| PROJECT_REPORT.md       | 19.6 KB  | The complete technical report in Markdown: monitored machine, objectives, dataset, features, the prediction-horizon decision, results, inference engine, API, interface, verification, lessons, limitations and future scope. |
| README.md               | 8.9 KB   | Start here. Quick start, what each screen shows, requirements, how to rebuild everything, tests, project layout, key results, how to use your own data, troubleshooting.                                                      |
| requirements-render.txt | 562 B    | Slim list of only the libraries the web app needs when hosted on Render (faster cloud build).                                                                                                                                 |
| requirements.txt        | 1.0 KB   | Python libraries the project needs. Install with <i>pip install -r requirements.txt</i> .                                                                                                                                     |
| run_phase1.py           | 2.5 KB   | Runs the whole data-and-model pipeline in order: download dataset, expand it, EDA, train, evaluate. Options <i>--from N</i> and <i>--only N</i> run part of it.                                                               |

### 3. src/ - data and model pipeline (Phase 1)

Everything that turns the raw dataset into a trained, evaluated Random Forest. Run the whole chain with `run_phase1.py`.

#### src/

| File                       | Size   | What it is                                                                                                                                                                                                                              |
|----------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>__init__.py</code>   | 0 B    | Marks <code>src/</code> as a Python package. Empty.                                                                                                                                                                                     |
| <code>config.py</code>     | 9.4 KB | Single source of truth for the whole project: the machine name (CNC Milling Machine), file paths, the five-sensor schema and limits, failure-mode causes and maintenance actions, physics thresholds, alarm bands and the colour theme. |
| <code>plot_style.py</code> | 2.0 KB | Shared dark 'control room' style for every matplotlib chart, so report figures match the web interface.                                                                                                                                 |

#### src/data/

| File                             | Size    | What it is                                                                                                                                                                                                                   |
|----------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>__init__.py</code>         | 0 B     | Package marker. Empty.                                                                                                                                                                                                       |
| <code>download_dataset.py</code> | 4.9 KB  | Step 1. Downloads the real AI4I 2020 dataset (10,000 rows) from the UCI repository into <code>data/raw/</code> .                                                                                                             |
| <code>eda.py</code>              | 19.2 KB | Step 3. Exploratory data analysis: draws figures 01-08 and writes <code>eda_summary.json</code> .                                                                                                                            |
| <code>expand_dataset.py</code>   | 16.1 KB | Step 2. Turns the 10,000 independent snapshots into run-to-failure histories for 220 simulated milling machines (74,979 rows), using the exact failure rules of the real data. Adds machine ID, cycle, time, RUL and labels. |

#### src/evaluation/

| File                          | Size    | What it is                                                                                                                                                                                              |
|-------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>__init__.py</code>      | 0 B     | Package marker. Empty.                                                                                                                                                                                  |
| <code>alarm_quality.py</code> | 3.8 KB  | Checks whether the model's 'false alarms' are really early warnings on machines that do fail later. Writes <code>false_positive_analysis.json</code> .                                                  |
| <code>evaluate.py</code>      | 17.2 KB | Step 6. Tests the model on 42 machines it never saw: confusion matrix, ROC/PR curves, feature importance, learning curve, early-warning lead time (figures 09-14) and <code>model_metrics.json</code> . |

#### src/features/

| File                           | Size   | What it is                                                                                                                                                                            |
|--------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>__init__.py</code>       | 0 B    | Package marker. Empty.                                                                                                                                                                |
| <code>build_features.py</code> | 8.0 KB | Step 4. Turns the 5 raw sensor readings into the 10 model features (temperature difference, power, strain...). Used by BOTH training and the live app, so the two can never disagree. |

#### src/models/

| File           | Size    | What it is                                                                                                                                           |
|----------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| __init__.py    | 0 B     | Package marker. Empty.                                                                                                                               |
| train_model.py | 13.2 KB | Step 5. Trains the Random Forest: grouped cross-validated hyper-parameter search, then a refit on the full training set. Saves the model to models/. |

## 4. backend/ - engine and API (Phase 2)

The server: prediction, health, RUL, alarms, explanations, storage and the REST API.

### backend/

| File        | Size   | What it is                                                                                                                                                                   |
|-------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| __init__.py | 0 B    | Package marker. Empty.                                                                                                                                                       |
| app.py      | 5.3 KB | Starts the application: one FastAPI server that serves the web interface (/), the REST API (/api), API docs (/docs) and the figures. Run with <i>python -m backend.app</i> . |

### backend/api/

| File        | Size    | What it is                                                                                                                                                  |
|-------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| __init__.py | 0 B     | Package marker. Empty.                                                                                                                                      |
| routes.py   | 19.3 KB | Every REST endpoint the web interface uses: predict (single and batch upload), machines, live replay and its live stream, alarms, model information, reset. |
| schemas.py  | 3.8 KB  | Input/output formats and validation rules. Rejects impossible readings (e.g. negative speed) before they reach the model.                                   |

### backend/core/

| File        | Size    | What it is                                                                                                                                                      |
|-------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| __init__.py | 0 B     | Package marker. Empty.                                                                                                                                          |
| alarms.py   | 9.3 KB  | Alarm engine: NORMAL / ADVISORY / WARNING / CRITICAL, the worst of three signals wins, plus a persistence filter that stops flickering alarms.                  |
| database.py | 17.3 KB | SQLite storage for machines, readings and the alarm log (the software stand-in for the synopsis's cloud store).                                                 |
| health.py   | 9.6 KB  | Calculates the 0-100 Health Index and the Remaining Useful Life (RUL) estimate, including the explanation of how the RUL was worked out.                        |
| physics.py  | 10.1 KB | Physics stress model: how close each reading is to each of the four failure limits (0 = safe, 1 = on the limit). Independent check alongside the Random Forest. |

### backend/services/

| File         | Size    | What it is                                                                                                                                                  |
|--------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| __init__.py  | 0 B     | Package marker. Empty.                                                                                                                                      |
| explainer.py | 9.1 KB  | Explains each individual prediction exactly (decision-path attribution): which sensor pushed the model toward its answer, and by how much.                  |
| predictor.py | 17.7 KB | The inference orchestrator: one reading in, full verdict out (class, probabilities, health, RUL, alarm, action, explanation). Everything goes through here. |
| simulator.py | 12.9 KB | Replays a real run-to-failure history one reading at a time, replacing the synopsis's NodeMCU sensor rig. Drives the 'Start Demo Plant' button.             |



## 5. frontend/ - the web interface (Phase 3)

The SCADA-style operator screens. Plain HTML/CSS/JavaScript, no build step, served by the backend.

### frontend/css/

| File      | Size    | What it is                                                                              |
|-----------|---------|-----------------------------------------------------------------------------------------|
| style.css | 17.6 KB | All styling: dark SCADA control-room theme, cards, gauges, alarm colours. No framework. |

### frontend/icons/

| File                  | Size    | What it is                                                     |
|-----------------------|---------|----------------------------------------------------------------|
| apple-touch-icon.png  | 11.7 KB | Home-screen icon for iPhone and iPad.                          |
| icon-192.png          | 12.3 KB | App icon (192 px) used by the installed PWA.                   |
| icon-512.png          | 30.9 KB | App icon (512 px) used on splash screens and stores.           |
| icon-maskable-512.png | 25.6 KB | Padded app icon Android crops into a circle or rounded square. |

### frontend/

| File          | Size   | What it is                                                                                                        |
|---------------|--------|-------------------------------------------------------------------------------------------------------------------|
| index.html    | 4.0 KB | The single web page. Top bar (with 'CNC Milling Machine Monitoring'), the six tabs, and the script loading order. |
| manifest.json | 749 B  | PWA manifest: app name, icons and colours so the browser can install the HMI as a desktop/phone app.              |
| sw.js         | 2.3 KB | PWA service worker: caches the page files so the app opens offline; live /api data is never cached.               |

### frontend/js/

| File       | Size    | What it is                                                                                                         |
|------------|---------|--------------------------------------------------------------------------------------------------------------------|
| api.js     | 5.0 KB  | Talks to the backend API and turns server errors into readable messages.                                           |
| app.js     | 6.1 KB  | Application shell: switches between screens, starts and stops each one, shared helpers.                            |
| pwa.js     | 4.4 KB  | Registers the service worker and shows the 'Install PdM SCADA' popup and top-bar Install button.                   |
| widgets.js | 17.7 KB | Hand-drawn SVG gauges, health ring and trend charts, with no charting library, so the demo works without internet. |

### frontend/js/screens/

| File      | Size    | What it is                                                                                            |
|-----------|---------|-------------------------------------------------------------------------------------------------------|
| alarms.js | 6.8 KB  | Screen 5, Alarms: the maintenance log with filters and Acknowledge.                                   |
| batch.js  | 10.9 KB | Screen 4, Batch Analysis: upload a CSV/Excel file, see every row scored, download the annotated file. |

| File        | Size    | What it is                                                                                                              |
|-------------|---------|-------------------------------------------------------------------------------------------------------------------------|
| machine.js  | 18.8 KB | Screen 2, Machine HMI: five sensor dials, health ring, RUL, stress bars, live trends, alarm banner, START/STOP.         |
| manual.js   | 12.8 KB | Screen 3, Manual Input: type or slide in one reading, or pick a preset, and get the full verdict.                       |
| model.js    | 13.5 KB | Screen 6, Model Insights: lead time, metrics, confusion matrix, feature importance and the live explanation self-check. |
| overview.js | 7.8 KB  | Screen 1, Plant Overview: one status card per machine, worst first; Start Demo / Stop All / Reset.                      |

## 6. data/ - datasets and the live database

### data/

| File        | Size     | What it is                                                                                                                           |
|-------------|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| pdm.sqlite3 | 348.0 KB | The live application database: registered machines, every reading and the alarm log. Created on first run; 'Reset Plant' empties it. |

### data/processed/

| File                     | Size   | What it is                                                                                                                                  |
|--------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------|
| pdm_expanded_dataset.csv | 7.5 MB | The training dataset: 74,979 rows = 10,000 real + 64,979 simulated from 220 run-to-failure machines. Rebuilt (seeded) by expand_dataset.py. |

### data/raw/

| File         | Size     | What it is                                                                                                |
|--------------|----------|-----------------------------------------------------------------------------------------------------------|
| ai4i2020.csv | 511.6 KB | The real UCI AI4I 2020 dataset: 10,000 rows, 339 failures. Re-download with src/data/download_dataset.py. |

## 7. models/ - trained model

Produced by `src/models/train_model.py`.

| File                                  | Size    | What it is                                                                                                                |
|---------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------|
| <code>pdm_model_bundle.joblib</code>  | 10.8 MB | The model the app actually loads: the trained Random Forest plus its feature list, class names and settings, in one file. |
| <code>random_forest_pdm.joblib</code> | 10.8 MB | The trained Random Forest on its own (the bare estimator), kept for use outside the app.                                  |

## 8. artifacts/ - figures and metrics

Produced by eda.py, evaluate.py and alarm\_quality.py. Used by the reports and the Model Insights screen.

### artifacts/metrics/

| File                         | Size     | What it is                                                                                                                                           |
|------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| eda_summary.json             | 2.4 KB   | Numbers from the data analysis: class counts, imbalance ratio, sensor statistics.                                                                    |
| false_positive_analysis.json | 242 B    | Output of alarm_quality.py: how many 'false alarms' were really early warnings.                                                                      |
| model_metrics.json           | 4.1 KB   | All evaluation results: accuracy, F1, per-class scores, ROC-AUC, early-warning lead time (8.67 h, 97.6 % warned). Read by the Model Insights screen. |
| split_indices.npz            | 111.6 KB | The exact train/test row split (grouped by machine), so evaluation uses the same unseen machines every time.                                         |
| train_log.txt                | 1.4 KB   | Console log of the last training run.                                                                                                                |
| training_summary.json        | 2.2 KB   | Training record: best hyper-parameters, search scores, run times, split sizes.                                                                       |

### artifacts/plots/eda/

| File                           | Size     | What it is                                                        |
|--------------------------------|----------|-------------------------------------------------------------------|
| 01_class_distribution.png      | 65.3 KB  | How many readings of each class; shows the 138:1 imbalance.       |
| 02_sensor_distributions.png    | 170.9 KB | Each sensor's spread, per failure class.                          |
| 03_correlation_matrix.png      | 116.1 KB | Correlation between all sensors and features.                     |
| 04_physics_separation.png      | 779.0 KB | Each failure mode sits inside its physical limit zone.            |
| 05_failure_by_machine_type.png | 70.5 KB  | Failure rate and mix for product grades L / M / H.                |
| 06_degradation_signatures.png  | 501.0 KB | How the key quantity drifts toward its limit before each failure. |
| 07_rul_distribution.png        | 66.0 KB  | Remaining-useful-life and machine life distributions.             |
| 08_boxplots_by_class.png       | 135.4 KB | Box plots of every feature per failure class.                     |

### artifacts/plots/evaluation/

| File                      | Size     | What it is                                                                 |
|---------------------------|----------|----------------------------------------------------------------------------|
| 09_confusion_matrix.png   | 75.8 KB  | Confusion matrix on the held-out test machines.                            |
| 10_roc_and_pr_curves.png  | 165.4 KB | ROC and precision-recall curves per failure mode.                          |
| 12_feature_importance.png | 91.7 KB  | Which features drive the prediction (impurity and permutation importance). |
| 13_learning_curve.png     | 66.4 KB  | Score versus training-set size: no overfitting, not data-starved.          |

| File                           | Size    | What it is                                                        |
|--------------------------------|---------|-------------------------------------------------------------------|
| 14_early_warning_lead_time.png | 76.6 KB | How many hours before failure the warning came, per failure mode. |

## 9. docs/ - documents and their generators

The PDFs are generated from a recorded run of the real app, never typed by hand. See docs/README.md for how to rebuild them.

### docs/

| File                                  | Size    | What it is                                                                                                                          |
|---------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------|
| 01_TEST_CASES_AND_EXPECTED_OUTPUT.pdf | 3.4 MB  | Full test document: 159 test cases with exact expected values, screen by screen.                                                    |
| 02_FINAL_YEAR_PROJECT_REPORT.pdf      | 2.2 MB  | The academic project report: certificate, declaration, abstract, 8 chapters, references, appendices.                                |
| 03_USER_INTERFACE_GUIDE.pdf           | 4.0 MB  | Plain-English guide to every screen, for readers with no technical background.                                                      |
| 04_PROJECT_FILE_GUIDE.pdf             | 54.6 KB | This document.                                                                                                                      |
| capture_evidence.py                   | 11.8 KB | Drives the real app in Chromium: takes the 16 screenshots and records evidence.json. Fails if any screen has a JavaScript error.    |
| diagrams.py                           | 8.2 KB  | Draws the vector block diagrams used in the project report.                                                                         |
| ESSENTIAL_WEBSITE_TEST_CASES.md       | 4.2 KB  | The 12 must-pass website tests for a demo or viva, with exact expected results and a pass column.                                   |
| evidence.json                         | 92.6 KB | Recorded outputs of a real browser run (API responses, predictions, validation errors). Every number in the PDFs is read from here. |
| gen_file_guide.py                     | 29.2 KB | Builds this document (04_PROJECT_FILE_GUIDE.pdf) by walking the project folder.                                                     |
| gen_project_report.py                 | 70.7 KB | Builds 02_FINAL_YEAR_PROJECT_REPORT.pdf.                                                                                            |
| gen_test_report.py                    | 56.1 KB | Builds 01_TEST_CASES_AND_EXPECTED_OUTPUT.pdf.                                                                                       |
| gen_ui_guide.py                       | 51.3 KB | Builds 03_USER_INTERFACE_GUIDE.pdf.                                                                                                 |
| machine_capture.py                    | 8.2 KB  | Helper for capture_evidence.py: reliably captures the two Machine HMI states (warning predicted / limit breached).                  |
| pdf_kit.py                            | 18.0 KB | Shared PDF toolkit: page layout, styles, tables, callouts, cover page, figure trimming and cropping.                                |
| README.md                             | 3.9 KB  | Explains the documents in docs/ and how to regenerate them.                                                                         |

### docs/screenshots/

| File                     | Size     | What it is                                              |
|--------------------------|----------|---------------------------------------------------------|
| 01_overview_empty.png    | 173.9 KB | Screenshot. Plant Overview before the demo starts.      |
| 02_overview_starting.png | 191.3 KB | Screenshot. Plant Overview just after Start Demo Plant. |
| 03_overview_running.png  | 240.4 KB | Screenshot. Plant Overview with four machines running.  |
| 04_machine_picker.png    | 213.8 KB | Screenshot. Machine HMI: choosing a machine.            |

| File                               | Size     | What it is                                                        |
|------------------------------------|----------|-------------------------------------------------------------------|
| 05_machine_hmi.png                 | 501.2 KB | Screenshot. Machine HMI running normally.                         |
| 06_machine_predicted.png           | 448.4 KB | Screenshot. Machine HMI with a failure PREDICTED (still running). |
| 07_machine_breached.png            | 491.1 KB | Screenshot. Machine HMI after the limit was BREACHED.             |
| 08_manual_healthy.png              | 461.8 KB | Screenshot. Manual Input: healthy preset.                         |
| 09_manual_hdf.png                  | 513.9 KB | Screenshot. Manual Input: heat-dissipation preset.                |
| 10_manual_pwf.png                  | 494.3 KB | Screenshot. Manual Input: power-failure preset.                   |
| 11_manual_osf.png                  | 512.7 KB | Screenshot. Manual Input: overstrain preset.                      |
| 12_manual_twf.png                  | 489.0 KB | Screenshot. Manual Input: tool-wear preset.                       |
| 13_batch_empty.png                 | 162.6 KB | Screenshot. Batch Analysis before upload.                         |
| 14_batch_results.png               | 291.2 KB | Screenshot. Batch Analysis with scored results.                   |
| 15_alarms.png                      | 592.7 KB | Screenshot. Alarms and maintenance log.                           |
| 16_model_insights.png              | 389.3 KB | Screenshot. Model Insights screen.                                |
| art_04_physics_separation.png      | 779.0 KB | Screenshot. Copy of EDA figure 04 used in the reports.            |
| art_09_confusion_matrix.png        | 75.8 KB  | Screenshot. Copy of figure 09 used in the reports.                |
| art_12_feature_importance.png      | 91.7 KB  | Screenshot. Copy of figure 12 used in the reports.                |
| art_14_early_warning_lead_time.png | 76.6 KB  | Screenshot. Copy of figure 14 used in the reports.                |

## 10. tests/ - automated tests

Run with `python -m pytest tests/ -q`, `node tests/test_widgets.js` and `node tests/test_boot.js`.

| File                         | Size    | What it is                                                                                                      |
|------------------------------|---------|-----------------------------------------------------------------------------------------------------------------|
| <code>__init__.py</code>     | 0 B     | Package marker. Empty.                                                                                          |
| <code>test_api.py</code>     | 12.0 KB | API tests (28) run on a temporary database: endpoints, validation, batch upload, alarms and reset.              |
| <code>test_boot.js</code>    | 5.8 KB  | Checks the page loads its scripts in a working order (guards a real bug that once blanked the whole interface). |
| <code>test_engine.py</code>  | 11.5 KB | Tests for physics, health, alarms and the explainer, including regression tests for bugs fixed in Phase 2.      |
| <code>test_widgets.js</code> | 4.2 KB  | Checks the SVG gauges with extreme and missing values (no NaN / undefined output). Run with <code>node</code> . |

## 11. batch\_testing/ - Excel files for the Batch Analysis screen

Ready-made files to upload on the Batch Analysis screen, grouped by what they test. Descriptions below come from 00\_TEST\_INDEX.xlsx. Rebuild the folder with generate\_batch\_tests.py.

### batch\_testing/

| File               | Size    | What it is                                                                                                                                                                    |
|--------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00_TEST_INDEX.xlsx | 19.9 KB | Index of every batch-test file: what it tests, the expected behaviour, and what the app actually returned (HTTP status, rows scored, classes, alarm levels). Open this first. |

### batch\_testing/01\_Healthy\_Machines/Mixed\_Types/

| File                               | Size   | What it is                                                                                                     |
|------------------------------------|--------|----------------------------------------------------------------------------------------------------------------|
| healthy_snapshot_100_machines.xlsx | 8.8 KB | 100 unrelated healthy machines, one reading each, mixed L/M/H. <b>Expected:</b> All or almost all rows Normal. |

### batch\_testing/01\_Healthy\_Machines/Type\_H\_High\_Quality/

| File                              | Size   | What it is                                                                                                                                              |
|-----------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| H_fleet_5_machines.xlsx           | 9.2 KB | Five healthy type-H machines at different stable operating points, 25 readings each. <b>Expected:</b> All rows Normal; summary reports 5 machines.      |
| H_single_machine_60_readings.xlsx | 7.0 KB | One healthy type-H machine, 60 readings, all sensors in the safe envelope, wear 5->90 min. <b>Expected:</b> All rows Normal, alarm NORMAL, high health. |

### batch\_testing/01\_Healthy\_Machines/Type\_L\_Low\_Quality/

| File                              | Size   | What it is                                                                                                                                              |
|-----------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| L_fleet_5_machines.xlsx           | 9.2 KB | Five healthy type-L machines at different stable operating points, 25 readings each. <b>Expected:</b> All rows Normal; summary reports 5 machines.      |
| L_single_machine_60_readings.xlsx | 7.0 KB | One healthy type-L machine, 60 readings, all sensors in the safe envelope, wear 5->90 min. <b>Expected:</b> All rows Normal, alarm NORMAL, high health. |

### batch\_testing/01\_Healthy\_Machines/Type\_M\_Medium\_Quality/

| File                              | Size   | What it is                                                                                                                                              |
|-----------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| M_fleet_5_machines.xlsx           | 9.2 KB | Five healthy type-M machines at different stable operating points, 25 readings each. <b>Expected:</b> All rows Normal; summary reports 5 machines.      |
| M_single_machine_60_readings.xlsx | 7.0 KB | One healthy type-M machine, 60 readings, all sensors in the safe envelope, wear 5->90 min. <b>Expected:</b> All rows Normal, alarm NORMAL, high health. |

### batch\_testing/02\_Failure\_Modes\_Synthetic/HDF\_Heat\_Dissipation/

| File                            | Size   | What it is                                                                                                                                                                                      |
|---------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HDF_confirmed_failure.xlsx      | 6.1 KB | Temp delta 8.3->7.6 K with speed 1360->1320 rpm: inside the HDF zone on every row. <b>Expected:</b> Rows predicted HDF, alarm CRITICAL.                                                         |
| HDF_early_warning_approach.xlsx | 6.4 KB | Temp delta falls 10.4->8.9 K and speed 1500->1390 rpm: approaching but not crossing the HDF limit. <b>Expected:</b> Health declines through the run; later rows may raise an early HDF warning. |
| HDF_full_degradation_run.xlsx   | 9.0 KB | Healthy start degrading all the way into HDF over 120 readings. <b>Expected:</b> Normal at the start, HDF by the end, alarms escalate.                                                          |

### batch\_testing/02\_Failure\_Modes\_Synthetic/OSF\_Overstrain/

| File                                    | Size   | What it is                                                                                                                                                                                                                                                                      |
|-----------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OSF_same_readings_all_three_type s.xlsx | 6.6 KB | Identical readings (torque 60, wear 185->210) on L, M and H machines - strain 11100->12600. <b>Expected:</b> Model flags OSF on all three, but hard breaches differ by type: L on every row, M only once strain passes 12000, H never (limit 13000) - shows the per-type limit. |
| OSF_type_H_limit_13000.xlsx             | 6.8 KB | Type H: tool wear x torque climbs past the 13000 min-Nm overstrain limit (power kept under 9000 W). <b>Expected:</b> Early rows Normal; final rows predicted OSF with CRITICAL alarm and hard breaches.                                                                         |
| OSF_type_L_limit_11000.xlsx             | 6.7 KB | Type L: tool wear x torque climbs past the 11000 min-Nm overstrain limit (power kept under 9000 W). <b>Expected:</b> Early rows Normal; final rows predicted OSF with CRITICAL alarm and hard breaches.                                                                         |
| OSF_type_M_limit_12000.xlsx             | 6.7 KB | Type M: tool wear x torque climbs past the 12000 min-Nm overstrain limit (power kept under 9000 W). <b>Expected:</b> Early rows Normal; final rows predicted OSF with CRITICAL alarm and hard breaches.                                                                         |

### batch\_testing/02\_Failure\_Modes\_Synthetic/PWF\_Power\_Failure/

| File                              | Size   | What it is                                                                                                                  |
|-----------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------|
| PWF_overload_degradation_run.xlsx | 8.9 KB | Torque climbs 40->66 Nm over 120 readings, pushing power past 9000 W. <b>Expected:</b> Normal at the start, PWF by the end. |
| PWF_overload_high_power.xlsx      | 6.1 KB | Torque 62->66 Nm at ~1450 rpm: mechanical power above 9000 W. <b>Expected:</b> Rows predicted PWF, alarm CRITICAL.          |
| PWF_underload_low_power.xlsx      | 6.1 KB | Torque 12->10 Nm at ~2650 rpm: mechanical power below 3500 W. <b>Expected:</b> Rows predicted PWF, alarm CRITICAL.          |

### batch\_testing/02\_Failure\_Modes\_Synthetic/TWF\_Tool\_Wear/

| File                        | Size   | What it is                                                                                                                                                                  |
|-----------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TWF_inside_wear_window.xlsx | 6.5 KB | Every reading inside the 200-240 min tool-wear window at a light load. <b>Expected:</b> Mostly TWF predictions (tool replacement due).                                      |
| TWF_tool_change_reset.xlsx  | 7.0 KB | Wear climbs 180->236 min, then the tool is changed and the counter resets to 0. <b>Expected:</b> TWF risk near the end of the first half, back to Normal after the reset.   |
| TWF_wear_ramp_0_to_250.xlsx | 9.1 KB | Tool wear counts 0->250 min at a normal load, every other sensor healthy. <b>Expected:</b> Normal while wear is low; TWF / maintenance advisory as wear enters 200-240 min. |

### batch\_testing/03\_Run\_To\_Failure\_Real\_Dataset/All\_Modes\_Combined/

| File                            | Size    | What it is                                                                                                                                                 |
|---------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| five_machines_one_per_mode.xlsx | 62.5 KB | One full run-to-failure machine per failure mode stacked in one file. <b>Expected:</b> Summary shows 5 machines with a mix of HDF/PWF/OSF/TWF predictions. |

### batch\_testing/03\_Run\_To\_Failure\_Real\_Dataset/HDF\_Heat\_Dissipation/

| File                            | Size    | What it is                                                                                                                                                                                                                 |
|---------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MC-009_type_L_188_readings.xlsx | 15.3 KB | Complete simulated life of MC-009 ending in HDF. 'actual_*' columns are ground truth (ignored by the app). <b>Expected:</b> Health falls over the run; HDF predicted before/at the failure. Compare with actual_* columns. |
| MC-023_type_L_235_readings.xlsx | 17.7 KB | Complete simulated life of MC-023 ending in HDF. 'actual_*' columns are ground truth (ignored by the app). <b>Expected:</b> Health falls over the run; HDF predicted before/at the failure. Compare with actual_* columns. |

### batch\_testing/03\_Run\_To\_Failure\_Real\_Dataset/OSF\_Overstrain/

| File                            | Size    | What it is                                                                                                                                                                                                                 |
|---------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MC-010_type_L_386_readings.xlsx | 25.8 KB | Complete simulated life of MC-010 ending in OSF. 'actual_*' columns are ground truth (ignored by the app). <b>Expected:</b> Health falls over the run; OSF predicted before/at the failure. Compare with actual_* columns. |
| MC-011_type_H_401_readings.xlsx | 26.5 KB | Complete simulated life of MC-011 ending in OSF. 'actual_*' columns are ground truth (ignored by the app). <b>Expected:</b> Health falls over the run; OSF predicted before/at the failure. Compare with actual_* columns. |

### batch\_testing/03\_Run\_To\_Failure\_Real\_Dataset/PWF\_Overload/

| File                            | Size    | What it is                                                                                                                                                                                                                 |
|---------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MC-001_type_M_359_readings.xlsx | 24.2 KB | Complete simulated life of MC-001 ending in PWF. 'actual_*' columns are ground truth (ignored by the app). <b>Expected:</b> Health falls over the run; PWF predicted before/at the failure. Compare with actual_* columns. |
| MC-024_type_H_323_readings.xlsx | 22.0 KB | Complete simulated life of MC-024 ending in PWF. 'actual_*' columns are ground truth (ignored by the app). <b>Expected:</b> Health falls over the run; PWF predicted before/at the failure. Compare with actual_* columns. |

### batch\_testing/03\_Run\_To\_Failure\_Real\_Dataset/PWF\_Underload/

| File                            | Size    | What it is                                                                                                                                                                                                                 |
|---------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MC-026_type_L_394_readings.xlsx | 25.9 KB | Complete simulated life of MC-026 ending in PWF. 'actual_*' columns are ground truth (ignored by the app). <b>Expected:</b> Health falls over the run; PWF predicted before/at the failure. Compare with actual_* columns. |
| MC-035_type_M_366_readings.xlsx | 24.4 KB | Complete simulated life of MC-035 ending in PWF. 'actual_*' columns are ground truth (ignored by the app). <b>Expected:</b> Health falls over the run; PWF predicted before/at the failure. Compare with actual_* columns. |

### batch\_testing/03\_Run\_To\_Failure\_Real\_Dataset/TWF\_Tool\_Wear/

| File                            | Size    | What it is                                                                                                                                                                                                                 |
|---------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MC-002_type_L_282_readings.xlsx | 20.0 KB | Complete simulated life of MC-002 ending in TWF. 'actual_*' columns are ground truth (ignored by the app). <b>Expected:</b> Health falls over the run; TWF predicted before/at the failure. Compare with actual_* columns. |
| MC-003_type_L_214_readings.xlsx | 16.6 KB | Complete simulated life of MC-003 ending in TWF. 'actual_*' columns are ground truth (ignored by the app). <b>Expected:</b> Health falls over the run; TWF predicted before/at the failure. Compare with actual_* columns. |

### batch\_testing/04\_Mixed\_Fleet/

| File                               | Size    | What it is                                                                                                                                                                                                 |
|------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| plant_8_machines_grouped.xlsx      | 15.8 KB | 8 machines on 3 lines, rows grouped by machine: 4 healthy, 1 HDF, 1 PWF, 1 OSF, 1 TWF trend. <b>Expected:</b> Summary: 8 machines, mix of Normal and failure classes; healthy lines stay NORMAL.           |
| plant_8_machines_interleaved.xlsx  | 16.2 KB | Same 8 machines but rows interleaved in time order (as a data logger would export them). <b>Expected:</b> Per-machine results identical to the grouped file - grouping by machine_id is order-independent. |
| shift_report_12_readings_each.xlsx | 8.4 KB  | Only the first 12 readings (one 2-hour shift) of each of the 8 machines. <b>Expected:</b> Degrading machines are still early; mostly Normal/ADVISORY.                                                      |

### batch\_testing/05\_Real\_AI4I\_2020\_Data/By\_Failure\_Flag/

| File                        | Size    | What it is                                                                                                                                      |
|-----------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ai4i_HDF_rows_115.xlsx      | 11.3 KB | The 115 real AI4I rows flagged HDF. <b>Expected:</b> Most rows predicted HDF (TWF is stochastic in AI4I, so lower agreement there is expected). |
| ai4i_normal_random_500.xlsx | 31.6 KB | 500 random real rows with no failure flag. <b>Expected:</b> Nearly all rows Normal.                                                             |
| ai4i_OSF_rows_98.xlsx       | 10.6 KB | The 98 real AI4I rows flagged OSF. <b>Expected:</b> Most rows predicted OSF (TWF is stochastic in AI4I, so lower agreement there is expected).  |
| ai4i_PWF_rows_95.xlsx       | 10.5 KB | The 95 real AI4I rows flagged PWF. <b>Expected:</b> Most rows predicted PWF (TWF is stochastic in AI4I, so lower agreement there is expected).  |
| ai4i_TWF_rows_46.xlsx       | 7.8 KB  | The 46 real AI4I rows flagged TWF. <b>Expected:</b> Most rows predicted TWF (TWF is stochastic in AI4I, so lower agreement there is expected).  |

### batch\_testing/05\_Real\_AI4I\_2020\_Data/By\_Machine\_Type/

| File                        | Size    | What it is                                                                                            |
|-----------------------------|---------|-------------------------------------------------------------------------------------------------------|
| ai4i_type_H_random_300.xlsx | 21.0 KB | 300 random real rows of product type H. <b>Expected:</b> Mostly Normal; a few failures where flagged. |
| ai4i_type_L_random_300.xlsx | 21.0 KB | 300 random real rows of product type L. <b>Expected:</b> Mostly Normal; a few failures where flagged. |
| ai4i_type_M_random_300.xlsx | 21.0 KB | 300 random real rows of product type M. <b>Expected:</b> Mostly Normal; a few failures where flagged. |

### batch\_testing/05\_Real\_AI4I\_2020\_Data/Original\_UCI\_Columns/

| File                       | Size    | What it is                                                                                                                                                                                                                                              |
|----------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ai4i_all_339_failures.xlsx | 23.3 KB | All 339 real rows labelled 'Machine failure = 1', original column names and failure flags. <b>Expected:</b> Large share predicted as a failure class; compare with TWF/HDF/PWF/OSF flag columns.                                                        |
| ai4i_first_500_rows.xlsx   | 29.7 KB | First 500 rows of the real UCI AI4I 2020 file with its original column names (Product ID, Type, Air temperature [K] ...). The UDI column is removed - see 10_Known_App_Issues. <b>Expected:</b> Accepted via column aliases; mostly Normal predictions. |

## batch\_testing/06\_Boundary\_Edge\_Cases/

| File                                | Size   | What it is                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HDF_threshold_8.6K_and_1380rpm.xlsx | 5.3 KB | Readings placed exactly on and either side of the HDF limits (delta 8.6 K AND speed 1380 rpm). Extra 'note' column is ignored. <b>Expected:</b> Only rows with BOTH delta < 8.6 and speed < 1380 are hard HDF breaches.                                                                                                                     |
| OSF_threshold_per_type.xlsx         | 5.5 KB | Tool wear 200 min at 1300 rpm (power stays under 9000 W) with torque chosen so strain sits 100 below / exactly on / 100 above each type's limit. <b>Expected:</b> Model predicts OSF on all rows (they are all near the limit); only the 'above' row per type is a hard overstrain breach.                                                  |
| PWF_threshold_3500W_and_9000W.xlsx  | 5.4 KB | Torque set at 1500 rpm so mechanical power lands 0.5 W and 1 W either side of the 3500 W and 9000 W limits (exactly 3500.000 W cannot be written as torque x rpm). <b>Expected:</b> Model may still say Normal (1 W is invisible to it), but exactly 4 rows - 3499, 3499.5, 9000.5, 9001 W - are hard power breaches with a CRITICAL alarm. |
| sensor_range_extremes.xlsx          | 5.3 KB | Sensor values at the HMI minimum, maximum, beyond the range, and with long decimals. <b>Expected:</b> All 4 rows scored (no range rejection); extreme rows get failure/breach results.                                                                                                                                                      |
| TWF_wear_window_200_240.xlsx        | 5.4 KB | Tool wear at 0, 199, 200, 201, 220, 239, 240, 241, 253 min, everything else nominal. <b>Expected:</b> Risk rises inside/after the 200-240 window; wear 0 and 199 read Normal.                                                                                                                                                               |

## batch\_testing/07\_Column\_Format\_Variants/

| File                                   | Size   | What it is                                                                                                                                      |
|----------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| canonical_column_names.xlsx            | 5.8 KB | Exact column names from the app's template. <b>Expected:</b> 20 rows scored.                                                                    |
| columns_reordered.xlsx                 | 5.8 KB | Canonical names in reverse column order. <b>Expected:</b> 20 rows scored; order does not matter.                                                |
| custom_sheet_name.xlsx                 | 5.8 KB | Data on a single sheet named 'Plant Log March' instead of Sheet1. <b>Expected:</b> 20 rows scored (the first sheet is read regardless of name). |
| extra_unused_columns.xlsx              | 6.1 KB | Valid data plus unrelated columns (operator, shift, notes, humidity_pct). <b>Expected:</b> 20 rows scored; extras ignored.                      |
| machine_type_lowercase_and_spaces.xlsx | 5.8 KB | machine_type written as 'm', ' m ', 'l', 'h'. <b>Expected:</b> Normalised to M/L/H; 20 rows scored.                                             |
| machine_type_unknown_values.xlsx       | 5.8 KB | machine_type values the app does not know (X, Medium, blank, Q). <b>Expected:</b> Rows still scored; unknown types fall back to L.              |
| no_machine_id_column.xlsx              | 5.7 KB | machine_id column omitted (it is optional). <b>Expected:</b> 20 rows scored; summary reports 0 machines.                                        |

| File                         | Size   | What it is                                                                                                                                   |
|------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------|
| spreadsheet_short_names.xlsx | 5.7 KB | Informal spreadsheet headers (Machine, Variant, air_temp, process_temp, RPM, Torque, ToolWear). <b>Expected:</b> 20 rows scored via aliases. |
| uci_ai4i_column_names.xlsx   | 5.8 KB | UCI AI4I header names (Product ID, Type, Air temperature [K] ...). <b>Expected:</b> 20 rows scored via aliases.                              |
| uppercase_with_spaces.xlsx   | 5.8 KB | Headers in UPPER CASE with spaces (AIR TEMPERATURE, TOOL WEAR ...). <b>Expected:</b> 20 rows scored.                                         |

### batch\_testing/08\_Invalid\_Inputs/Expect\_Partial\_Rows\_Rejected/

| File                           | Size   | What it is                                                                                                                                                                    |
|--------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| empty_rows_in_middle.xlsx      | 5.4 KB | Two completely empty rows between valid readings. <b>Expected:</b> HTTP 200 - 10 scored, 2 rejected.                                                                          |
| some_text_and_blank_cells.xlsx | 5.4 KB | 10 rows; row 4 torque 'N/A', row 7 air temp 'error', row 9 tool wear blank (Excel row numbers). <b>Expected:</b> HTTP 200 - 7 rows scored, 3 rejected with their row numbers. |
| zero_and_negative_speed.xlsx   | 5.4 KB | Rows with rotational_speed 0 and -1500 rpm. <b>Expected:</b> HTTP 200 - 8 scored, 2 rejected as non-physical.                                                                 |

### batch\_testing/08\_Invalid\_Inputs/Expect\_Rejected\_File/

| File                             | Size   | What it is                                                                                                                                           |
|----------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| all_rows_non_numeric.xlsx        | 5.4 KB | Every torque cell is the text 'high'. <b>Expected:</b> HTTP 422 - No valid rows found.                                                               |
| corrupted_not_really_excel.xlsx  | 46 B   | A text file renamed to .xlsx. <b>Expected:</b> HTTP 422 - Could not parse the file.                                                                  |
| data_on_second_sheet.xlsx        | 5.8 KB | First sheet is a cover page; the readings are on sheet 2. <b>Expected:</b> HTTP 422 - only the first sheet is read, so required columns are missing. |
| headers_only_no_rows.xlsx        | 5.0 KB | Correct headers but zero data rows. <b>Expected:</b> HTTP 422 - No rows found in the file.                                                           |
| missing_all_sensor_columns.xlsx  | 5.1 KB | Only machine_id and machine_type present. <b>Expected:</b> HTTP 422 listing all five sensor columns.                                                 |
| missing_machine_type_column.xlsx | 5.4 KB | Required 'machine_type' column removed. <b>Expected:</b> HTTP 422 - missing required column(s): machine_type.                                        |
| missing_torque_column.xlsx       | 5.4 KB | Required 'torque' column removed. <b>Expected:</b> HTTP 422 - missing required column(s): torque.                                                    |

### batch\_testing/09\_Volume\_Performance/

| File                     | Size     | What it is                                                                                                                    |
|--------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------|
| rows_100.xlsx            | 8.7 KB   | First 100 readings of the run-to-failure dataset (several complete machines). <b>Expected:</b> HTTP 200 - 100 rows scored.    |
| rows_1000.xlsx           | 40.9 KB  | First 1000 readings of the run-to-failure dataset (several complete machines). <b>Expected:</b> HTTP 200 - 1000 rows scored.  |
| rows_20000_at_limit.xlsx | 718.8 KB | Exactly 20,000 rows - the maximum the upload accepts. <b>Expected:</b> HTTP 200 - 20,000 rows scored (slow: tens of seconds). |

| File                       | Size     | What it is                                                                                                                   |
|----------------------------|----------|------------------------------------------------------------------------------------------------------------------------------|
| rows_20001_over_limit.xlsx | 718.8 KB | 20,001 rows - one more than the limit. <b>Expected:</b> HTTP 413 - the limit is 20,000.                                      |
| rows_5000.xlsx             | 184.5 KB | First 5000 readings of the run-to-failure dataset (several complete machines). <b>Expected:</b> HTTP 200 - 5000 rows scored. |

## batch\_testing/10\_Known\_App\_Issues/

| File                                              | Size    | What it is                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ai4i_full_original_header_UDI_and_Product_ID.xlsx | 16.3 KB | APP ISSUE: the unmodified UCI header contains both 'UDI' and 'Product ID'. The app aliases both to machine_id, producing two machine_id columns. <b>Expected:</b> Should be accepted. CURRENTLY rejected with HTTP 422 'Grouper for machine_id not 1-dimensional'. Workaround: delete the UDI column.                        |
| short_names_with_spaces_Air_Temp.xlsx             | 5.7 KB  | APP ISSUE: headers 'Air Temp' / 'Process Temp' (with a space). The alias table has 'air_temp' but the lookup runs before spaces are converted to underscores. <b>Expected:</b> Should be accepted. CURRENTLY rejected with HTTP 422 missing air_temperature, process_temperature. Workaround: write air_temp / process_temp. |

## 12. Tool and cache folders

These folders are created by tools, not written as part of the project. Their contents are not listed. Only `.venv` is needed to run the app, and it can be recreated at any time.

| Folder                      | What it is                                                                                                                                                                                                                               |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>.venv/</code>         | Python virtual environment: the interpreter and every installed library (FastAPI, scikit-learn, pandas, ReportLab, Playwright...). Recreate with <code>py -3.12 -m venv .venv</code> then <code>pip install -r requirements.txt</code> . |
| <code>.pytest_cache/</code> | Created automatically by pytest to remember the last test run. Safe to delete.                                                                                                                                                           |
| <code>__pycache__/</code>   | Compiled Python bytecode, created automatically whenever a module is imported (appears inside several folders). Safe to delete.                                                                                                          |
| <code>.kilo/</code>         | Settings folder left by the Kilo Code editor extension. Not used by the project.                                                                                                                                                         |
| <code>reports/</code>       | Empty folder. Not used by the project.                                                                                                                                                                                                   |
| <code>_prepared/</code>     | Inside docs/: whitespace-trimmed and cropped copies of the screenshots, made automatically by <code>pdf_kit.py</code> while building the PDFs. Safe to delete; rebuilt on the next PDF build.                                            |